

Using the mosquito transcriptome to investigate the role of floral humidity as a foraging cue in insect pollinators

MSc Bioinformatics Research Project Proposal



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Abstract (words)

Mosquitoes are responsible for transmitting some of the world's deadliest infectious diseases, making them valuable models for studying insect biology, given the extensive research that has produced an abundance of publicly available datasets. They also occupy a unique ecological position, with females requiring blood meals for reproduction while both sexes rely on floral nectar as their primary energy source. Host and oviposition site location are mediated by active humidity sensing, or hygrosensation, with ionotropic receptors Ir93a, Ir40a, and Ir68a, being well characterised in these contexts. Floral humidity has been demonstrated to act as a foraging cue for several insect pollinators, yet its role in mosquito foraging behaviour remains unexplored, and transcriptional dynamics of hygrosensory receptor expression across the mosquito life cycle are unknown. This project aims to determine whether floral humidity acts as a foraging cue for mosquitoes by examining hygrosensory receptor expression across sexes and life stages using transcriptomic data from *Aedes aegypti*. Males, as obligate nectar-feeders, provide a key comparison with females, whose feeding behaviour changes across the reproductive cycle. The analysis will subsequently be expanded across multiple mosquito species with contrasting ecological niches, enabling cross-species comparison of hygrosensory receptor expression and its relationship to foraging behaviour.

Keywords: mosquito, pollinator, foraging, floral humidity, hygrosensation

Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this proposal which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material, which is the work of others, I have included the source in the references. Any views expressed in the proposal, other than referenced material, are those of the author.



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1 Literature review (2,629 words)

1.1 Mosquitoes as vectors for disease

1.1.1 Mosquito-borne diseases and disease transmission

Mosquitoes are responsible for transmitting some of the deadliest diseases in human history, contributing substantially to the global burden of vector-borne disease (Lin et al., 2025; Naghavi et al., 2025). Additionally, the prevalence of these diseases continues to rise annually, with many mosquito species expanding their geographic range as a result of climate change and urbanisation (Carlson et al., 2023; Lim et al., 2025). *Aedes* mosquitoes such as *Ae. aegypti* and *Ae. albopictus* are principal vectors of four major arthropod-borne viruses (arboviruses), dengue, zika, chikungunya, and yellow fever, with dengue alone accounting for 14.1 million reported cases and 9,500 deaths in 2024 (Haider et al., 2025; Lim et al., 2025). Of particular concern is *Anopheles gambiae*, the dominant vector for the *Plasmodium falciparum* parasite which is responsible for the bulk of the malaria burden worldwide, estimated to have reached 282 million cases and 610,000 deaths in 2024 (WHO, 2025). This severe and far-reaching impact has resulted in mosquitoes receiving extensive research attention, producing an abundance of publicly available datasets, making them interesting candidates as models for examining insect biology and behaviour.

Disease transmission occurs via blood-feeding in female mosquitoes, a necessary stage in their reproductive cycle required for vitellogenesis and egg production, and although they evolved from generalist ancestors, many species are remarkably anthropophilic, having specialised for humans and human habitats (Bouafou et al., 2024; Crawford et al., 2017; Pierce et al., 2025; Wang et al., 2026). This specialisation extends to oviposition site, as well as host preference. *Ae. aegypti*, for example, tends to lay its eggs in small, water-holding containers such as tires and buckets, while *An. gambiae* prefers larger bodies of water, such as ponds and puddles, near human settlements (Bøgh et al., 2003; David et al., 2023; Munga et al., 2005; Ngugi et al., 2017; Wong et al., 2011). While these ecological proximities bring mosquitoes into frequent contact with potential hosts, successful blood meals require active host location.

1.1.2 Host location and humidity as a sensory cue

Hosts are located through a multimodal cascade of physical and chemical cues. Carbon dioxide (CO₂) from breath acts as a long-range indicator, followed by organic volatile odorants, then body heat, visual contrast, and humidity at shorter ranges (Coutinho-Abreu et al., 2022b; McMeniman et al., 2014; van Breugel et al., 2015; Zhou et al., 2018). Compared with other host-associated cues, humidity has historically received less attention, although recently it has been increasingly examined, with multiple studies demonstrating its importance in host-seeking and oviposition.

This dual requirement for humidity sensing makes mosquitoes particularly dependent on active hygrosensation (humidity sensing) throughout their gonotrophic cycle (Figure 1), and the hygrometers (humidity receptors) involved have been increasingly well characterised (Laursen et al., 2023; Tang et al., 2024; van Breugel et al., 2015).

1.2 Evolution of humidity sensing in insects

1.2.1 Hygrosensor architecture and receptors

As arthropods, the risk of desiccation imposes a selection pressure that selects for active humidity sensing (hygrosensation), allowing insects to detect and move along humidity gradients to avoid water loss (Enjin, 2017). Hygrosensory sensilla, located within insect antennae, have been well studied in a number of models, such as *Carausius morosus* (stick insect), *Periplaneta americana* (cockroach), *Drosophila melanogaster* (vinegar fly), and multiple species of mosquito. Sensilla typically contain paired moist and dry cells (Figure 2) which respond oppositely to changes in humidity, providing both directional and absolute hygrosensory information (Enjin et al., 2016; Knecht et al., 2017; Laursen et al., 2023; Tichy, 2007; Tichy and Kallina, 2010). The specific humidity receptors (hygrometers) associated with hygrosensory neurons have been identified as ionotropic receptors, with Ir40a typically associated with dry cells, Ir68a with moist, and Ir93a with both, acting as a co-receptor with the other two receptors in either cell type. Ir93a is also found in cold/thermoreceptor cells as well as olfactory neurons, indicating that it likely plays a wide-ranging role in mediating signal transduction across sensory inputs (Laursen et al., 2023).

1.2.2 Evolution of hygrometers

Interestingly, Ir93a and its partners are conserved from crustaceans to insects, with their origins estimated at a minimum of ~250 million years before the evolution of blood-feeding in *Culicidae*, which emerged ~100-200 million years ago (Pierce et al., 2025; Reidenbach et al., 2009). Because the crustacean ancestors of mosquitoes were aquatic and would have had no need to sense atmospheric humidity, these receptors likely evolved to perform a different ancestral function, such as mechanoreception or osmosensation, before then being co-opted for hygrosensation when early insects colonised land (Croset et al., 2010; Eyun et al., 2017; Kozma et al., 2018; Laursen et al., 2023). Consistent with this, mosquito hygrosensory neurons innervate poreless sensilla ampullaceous (Figure 2), which lack chemical access and are thought to detect humidity indirectly via humidity-dependent mechanical force, transmitted through the cuticle (Laursen et al., 2023).

Hygroreceptors in insects may have initially evolved out of the need to avoid desiccation, but the appearance of novel sensory apparatus can facilitate the development of novel behaviours. In mosquitoes, these receptors have been shown to be essential to host-seeking and oviposition site-location. However, research into their role in non-gonotrophic behaviours, such as nectar-feeding, is lacking (Laursen et al., 2023; Tang et al., 2024; van Breugel et al., 2015). This presents a significant gap, given the role of floral humidity as a foraging cue in other insects (Dahake et al., 2022; De Carvalho et al., 2021; Harrap et al., 2021; von Arx et al., 2012).

1.3 Floral humidity and mosquito foraging preferences

1.3.1 Floral humidity

Angiosperms are known to produce regions of elevated relative humidity around their floral displays. In a survey by Harrap et al. (2020), it was found that 30 out of the 42 flowering plant species examined (71%) produced floral humidity 1-4% above levels explicable by background environmental variation with gradients extending several centimetres from the corolla, supporting the idea that it is a genuine floral trait, as opposed to incidental evaporation. Floral humidity primarily arises from two mechanisms, namely nectar evaporation and petal/stomatal transpiration, with the relative contribution of each varying by species, indicating that different plants invest in either trait to different degrees as they evolve (Harrap and Rands, 2022). Further to this, Ke et al. (2024) examined 29 floral traits in 24 flowering species, finding that some flowering species appear to have invested in less conservative water usage strategies, increasing efficiency of water transpiration and size of floral displays despite increased susceptibility to drought conditions. This indicates that these plants gain some other benefit from such evolutionary adaptations, with increased attractiveness to pollinators being the most likely candidate. The greater water investment seen in basal angiosperms relative to monocots and eudicots is consistent with floral humidity being an ancestral trait subsequently modified across angiosperm lineages, although this would require phylogenetic analysis to confirm, a difficult task given the multiple traits that contribute to humidity production (Ke et al., 2024).

1.3.2 Signals, cues, exploitation and deception

Floral humidity has been shown to play a role in nectar foraging with a number of insect pollinators, including multiple hawkmoths and bumblebee species, preferentially visiting more humid flowers, but whether this is a cue or a signal is not always clear, and may vary on a case-by-case basis (Dahake et al., 2022; Harrap et al., 2021; von Arx et al., 2012). A cue constitutes information inadvertently produced, such as CO₂ from a respiring host or plant, whereas a signal refers to information whose production is favoured by selection because both sender and receiver

benefit, however the distinction between them is not always clear (Laidre and Johnstone, 2013). A trait may even originate as a cue and become more signal-like through selection. This is illustrated by Omondi et al. (2026), who found that *Lantana camara* CO₂ emissions were positively correlated with nectar production and response from *An. coluzzii* and *An. arabiensis* mosquitoes was dose-dependent, demonstrating a true signal-receiver system that had its probable origin as a cue (Schiestl and Johnson, 2013). Signals can also be selected for in a process of co-evolution, whereby a signalling-plant and receiver-pollinator might evolve in tandem, but this is often not the case. Instead, there are far more instances of angiosperms evolving traits that exploit pre-existing biases of insect pollinators, including chemical-communication systems and colourful displays (Schiestl and Johnson, 2013). Given that hygrosensation likely pre-dates angiosperm floral humidity, plants producing elevated floral humidity to attract pollinators are most likely tapping into insects' innate preference for humid environments, as opposed to producing a co-evolved mutual signal.

Importantly, traits that increase floral humidity may also have evolved for reasons other than attracting insect pollinators (Harrap et al., 2020; Harrap and Rands, 2022; Ke et al., 2024). Maintaining elevated humidity can help protect pollen from desiccation and increase the chance of successful germination, although this is a delicate balance to maintain, as the pollen can burst under turgor pressure if its water content becomes too high (Harrap et al., 2020; Von Hase et al., 2006). Additionally, transpirational cooling, which results in elevated humidity around the floral display, plays an important role in thermoregulation in high temperatures (Harrap and Rands, 2022; Huang et al., 2018; Sinha et al., 2022). Therefore, while floral humidity most likely functions primarily as a cue, the extent to which specific plants may have additionally evolved it as a signal remains difficult to isolate from these other selective pressures.

As previously indicated, many angiosperms produce floral humidity gradients to varying degrees, but the question remains as to whether there is a benefit to pollinators in visiting more humid flowers. If a plant were to produce humidity as a signal without any associated benefit, it would effectively be deceiving the visiting pollinator. Schiestl and Dötterl (2012) describe an instance of this, whereby several genera of *Araceae* have evolved to emit skatole, a compound used by scarabs to locate dung. Here, the plant deceives the beetle pollinators, attracting them by mimicking dung but offering nothing in return. There is some evidence to suggest that elevated humidity may result in less concentrated nectar due to a slower rate of evaporation (Brito Vera and Pérez, 2024; Chabert et al., 2020; Pacini and Nepi, 2007). If insects, such as mosquitoes, were found to have a preference for these flowers despite the lower nectar quality, it would strengthen the case for exploitation.

1.3.3 Mosquitoes as flower visitors and pollinators

Both male and female adult mosquitoes feed on floral nectar, requiring plant sugars as their primary energy source. Only females are haematophagous, requiring a blood meal to trigger vitellogenin synthesis and egg maturation, whereas males are obligate nectar-feeders (Day, 2016). Additionally, nectar-feeding predates blood-feeding evolutionarily, as ancestors of today's mosquitoes have been flower visitors since the Cretaceous period (Peach and Gries, 2020). Mosquitoes have been documented as pollinators in several systems, although notably, floral humidity has not been assessed in these instances despite its likely relevance, reflecting the gap in the literature regarding humidity's role in foraging by mosquitoes (Lahondère et al., 2020; Peach and Gries, 2016). This lack of research represents a significant lapse, given the importance of nectar-feeding to mosquito competence, longevity, and flight capacity (Foster, 2022; Upshur et al., 2019). In a recent review of mosquito-plant interactions, Foster (2024) concluded that most involve generalist visitation rather than specialist coevolution but acknowledged that multimodal cues (including floral humidity) are likely under explored in this system. Mosquitoes therefore provide a particularly useful model for studying floral humidity as a foraging cue, given that their feeding habits vary between sexes and, in the case of females, over the course of their life cycle, making intra-species comparisons possible, unlike most other pollinators.

1.4 Future work and perspectives

1.4.1 Hygroreceptor expression over the mosquito life cycle

While the role of hygrosensors in the mosquito gonotrophic cycle is well understood, their importance in nectar-feeding has yet to be explored (Laursen et al., 2023; Tang et al., 2024; van Breugel et al., 2015). To this end, it would be useful to examine transcriptional changes in the expression of key hygroreceptors (Ir93a, Ir40a, and Ir68a) in the female mosquito at different stages in their life cycle. Males, as obligate nectar-feeders throughout adult life, would provide a useful comparison (Day, 2016). If hygroreceptor expression is consistent in males but variable across female life stages, this would directly link expression to the female-specific behavioural transitions of the gonotrophic cycle. If expression remains relatively stable in females, this would be consistent with hygrosensation involvement in nectar-foraging, while not necessarily confirming it. Cross-species comparisons of hygroreceptor expression, in species with different ecological niches and behaviours, would help to elucidate the extent to which hygrosensation is involved. Additionally, comparing hygroreceptor expression with that of receptors of CO₂ and key volatile organic compounds (VOCs) might provide additional insight into their relative contributions, although importantly there may also be interplay between floral humidity and the mechanics of CO₂ and volatile dispersion and detection under different conditions, as these are

likely to behave differently at different temperatures and humidities (Coutinho-Abreu et al., 2022a; McMeniman et al., 2014; van Breugel et al., 2015; Zhou et al., 2018).

While transcriptomics analysis would allow for the simultaneous comparison of multiple receptors and may reveal other co-regulated receptors not previously implicated, it should be noted that transcript abundance only provides an approximation of protein expression. This remains a limitation of the approach and any findings would be strengthened by associated behavioural studies. It should also be noted that since *Ir93a* is found in other sensory cells besides hygrosensory moist and dry cells, any changes in *Ir93a* transcript abundance could reflect altered expression in other sensory apparatus associated with changes in behaviour (Laursen et al., 2023). Here, single-cell transcriptomic analysis would provide the capacity to differentiate between moist, dry, and thermoreceptor cells.

1.4.2 Comparison with other insect pollinators

Studying an organism whose feeding habits vary between sexes and, for females, at different stages of their life cycle, provides an exciting opportunity to better understand the role of hygrometers in foraging by insect pollinators more broadly. Ultimately, even if their expression in mosquitoes is relatively stable, this might simply indicate that the general need to avoid desiccation is greater in mosquitoes than any benefit conferred by visiting more humid flowers (Enjin, 2017). However, this may not be the case with other insects where the role of floral humidity as a cue has previously been investigated (Dahake et al., 2022; Harrap et al., 2021; von Arx et al., 2012). Despite multiple behavioural studies in this area, the mechanism behind such behaviours, and whether these specific hygrometers are involved, are not yet known. To understand the wider role of hygrometers in pollinator foraging, it would therefore be beneficial to compare the transcriptomes of mosquitoes to other insect pollinators. It should also be considered whether ambient humidity affects the relative importance of hygrometry. To investigate this, comparison of insects from varying humidity environments would be informative.

1.4.3 Phylogenetic analysis of hygrometry and floral humidity evolution

If these hygrometers are found to be involved in foraging and preference for flowers with elevated humidity, then it will be necessary to determine how these traits evolved in relation to each other. Phylogenetic dating of hygrometry diversification in insects, set against the timing of angiosperm radiation and floral humidity evolution, would reveal whether mosquito hygrosensory architecture predates the availability of floral humidity as a cue. If hygrometers substantially predate angiosperm floral humidity, this would be consistent with, and strongly support, the exploitation hypothesis, though it would not rule out subsequent co-evolutionary modification of these traits.

1.4.4 Project scope

The present study will examine hygrometric receptor expression throughout the mosquito life cycle across multiple species whose ecological differences include variation in foraging, host, and oviposition site preference. Comparison with other insect pollinators and phylogenetic analysis will be explored, time permitting.

2 Project proposal

2.1 Aims and objectives (242 words)

The primary aim of this project is to determine whether floral humidity acts as a foraging cue for mosquitoes by investigating the expression of key hygrometric receptors across their life cycle and between sexes. This aim can be broken down into two parts, whereby an initial pilot study will be performed, followed by an expanded study, with the commencement of the latter being dependent on the successful completion of the former. The objectives for each are as follows:

Pilot study:

- 1) Obtain *Ae. aegypti* RNA-Seq transcriptomic data.
- 2) Assemble *Ae. aegypti* transcriptome and quantify expression.
- 3) Translate transcriptome and reference genome to proteomes and identify protein families.
- 4) Compare expression of key hygrometric receptors between sexes over the *Ae. aegypti* life span.

Expanded study:

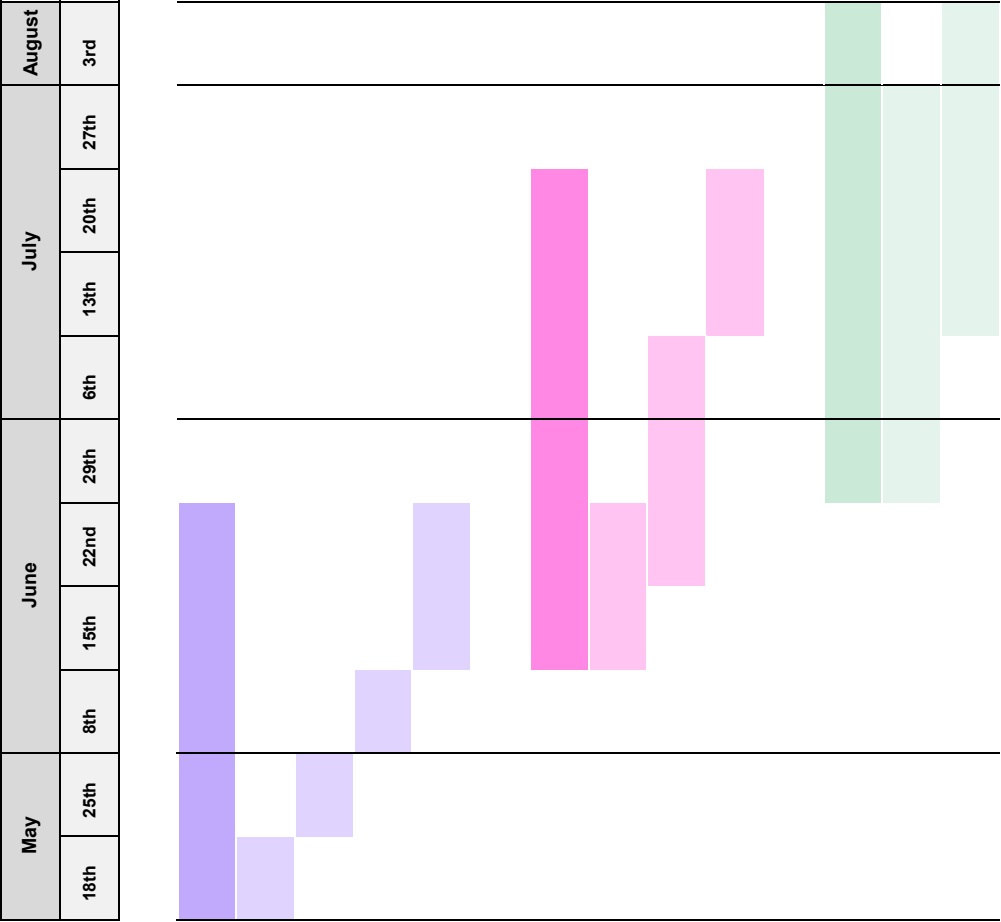
- 1) Expand transcriptomic dataset with additional mosquito species.
- 2) Assemble additional transcriptomes and quantify expression.
- 3) Translate transcriptomes and reference genomes to proteomes and identify protein families across species.
- 4) Normalise expression for each dataset.
- 5) Compare expression of hygrometric receptors across species with key ecological differences.
- 6) Compare expression of hygrometric receptors with that of volatile compound and CO₂ receptors.

If the primary aim is achieved, it will be possible to pursue additional lines of enquiry. Firstly, to investigate the importance of these receptors to insect pollinators more broadly, by comparing their expression in other insect species, and secondly, to determine whether floral humidity acts as a cue or signal, by timing its evolutionary emergence compared with that of insect hygro-sensation.

2.2 Plan (429 words)

The first stage of this project will be a pilot study with the goal of building an analysis pipeline to compare receptor expression. The purpose of this is to develop the core code necessary to perform the analysis and demonstrate the proof of concept. *Ae. aegypti*, along with *An. gambiae*, is possibly the most well studied mosquito species so has an abundance of publicly available, high quality well annotated transcriptomic datasets. *Ae. aegypti* transcriptomes will be downloaded from the NCBI Short Read Archive (SRA) using the SRA toolkit . Read quality will be assessed using FastQC and reads will be trimmed with fastp to remove any tags or low-quality base pairs. At this point, additional Quality Control (QC) and trimming software will be assessed to see whether they perform better. An *Ae. aegypti* reference genome will be retrieved and used to build HISAT2 indexes, mapping the trimmed reads to the reference genome for expression quantification, which will be performed using FeatureCounts. Protein families will be identified using OrthoFinder and the translated *Ae. aegypti* reference genome. The expression of key hygromet receptors will then be compared between male and female *Ae. aegypti* mosquitoes at different time points post emergence.

The second stage of the project will involve expanding the mosquito transcriptomic dataset, keeping track of species features, particularly any nectar-foraging, host, or oviposition site preference to see whether these are reflected in differential hygromet receptor expression between species. QC and trimming of these datasets will be performed, as before, and reference genomes will be retrieved where possible, for transcriptome assembly. The base code required for transcriptome assembly and quantification will have already been written at this point, so the task here will be to automate it to iterate over the additional datasets and troubleshoot any issues that arise. As part of this process, the putative list of species to be included in the final analysis will be finalised. The additional transcriptomes will be translated to proteomes, and protein families will be annotated using translated *Ae. aegypti*, *An. gambiae*, and *D. melanogaster* reference genomes. Again, the main task will be to automate as much of the process as possible. One aspect that makes comparing expression in multiple species difficult, is that they come from different sources so sequencing depth and library preparation will vary between datasets. To account for this, the datasets will have to be normalised before analysis, where expression of hygromet receptors will be compared between species along with expression of CO₂ receptors and receptors of key volatile organic compounds, as these are understood to play roles in the behaviours of interest.



TASK	START	END
Pilot study	18-May-2026	28-Jun-2026
Data collection, quality control and pre-processing	18-May-2026	24-May-2026
Genome assembly and quantification	25-May-2026	07-Jun-2026
Identification of protein families	08-Jun-2026	14-Jun-2026
Differential gene expression analysis (single species)	15-Jun-2026	28-Jun-2026
Expanded study	15-Jun-2026	16-Jul-2026
Expansion of transcriptomic dataset with additional species	15-Jun-2026	18-Jun-2026
Automation of pipeline across additional species datasets	22-Jun-2026	12-Jul-2026
Differential gene expression analysis (multiple species)	13-Jul-2026	26-Jul-2026
Data interpretation and write-up	29-Jun-2026	07-Aug-2026
Data interpretation	29-Jun-2026	02-Aug-2026
Write up	13-Jul-2026	07-Aug-2026

344 2.4 Risk register

Nature of the risk	Likelihood and potential impact on the project	Mitigation strategy / Contingency plan
Publicly available RNA-Seq datasets lack the metadata required for analysis (e.g. sex, life stage, or time point post-emergence are not recorded, or samples are not sex-separated).	Moderate-high likelihood. Would reduce the number of usable datasets and potentially limit the scope of life-stage or sex-based comparisons, particularly for less-studied species in the expanded study.	Dataset metadata will be assessed systematically during species selection before download. The pilot study will use <i>Ae. aegypti</i> , for which well-annotated, sex-separated datasets are known to exist. Species will only be included in the expanded study if suitable datasets are confirmed.
Reference genomes are unavailable for a subset of candidate mosquito species, preventing reference-based mapping.	Moderate likelihood given the breadth of species under consideration. Would require either cross-species mapping or exclusion of affected species, reducing the comparative power of the expanded study.	Species selection will prioritise those with published reference genomes. For species without a reference, mapping to a closely related species' genome will be attempted, with mapping rates assessed to determine usability. Species with consistently low mapping rates will be excluded.
Mapping rates are low due to poor reference genome quality or annotation.	Low-moderate likelihood for well-studied species, but higher for less-characterised ones.	Mapping rates will be checked for each sample, with those below an acceptable threshold being flagged. Reference genomes and annotation versions will be verified against the source database.
Datasets from different laboratories or sequencing runs introduce batch effects that confound cross-species comparisons.	Moderate-high likelihood given that datasets will be drawn from multiple sources with varying sequencing depth, library preparation, and platform. Could produce differential expression results unrelated to biology.	Normalisation will be applied to all datasets prior to cross-species comparison. Batch effects will be assessed and, where necessary, datasets will be removed.
The pilot study timeline is longer than anticipated due to difficulties with building code, using bioinformatics tools, or handling datasets.	Moderate likelihood , particularly with regard to unfamiliar analysis approaches and tools not previously used (OrthoFinder). Could delay the start of the expanded study.	Buffer time has been built into the project plan to allow for troubleshooting. If the pilot study takes longer than expected, the expanded study species list will be adjusted accordingly to maintain the planned timeline.
Differential expression analysis yields no significant changes in hygroreceptor expression across life stages or between sexes.	Low-moderate likelihood. Would limit the conclusions that can be drawn about the role of hygrosensation in nectar-foraging, though a negative result is itself informative.	The null result would be interpreted in the context of cross-species comparisons and existing behavioural literature. The analysis pipeline will also capture expression of CO ₂ and VOC receptors, providing comparative data.

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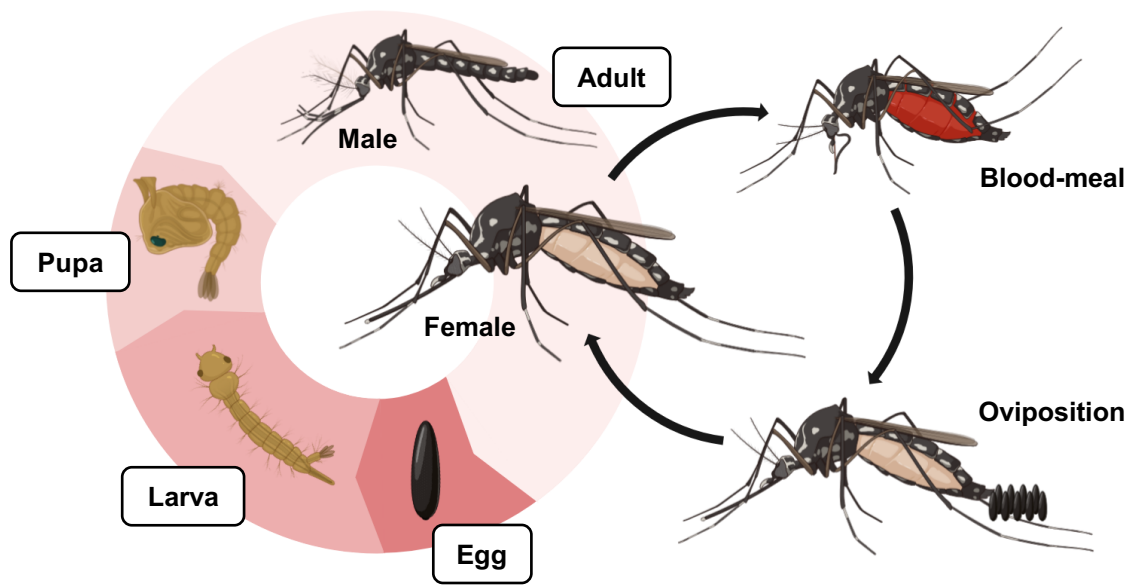


Figure 1. Mosquito life cycle. *Ae. aegypti* life cycle progressing from egg, to larva, to pupa to adult. Gonotrophic cycle is also shown for females.

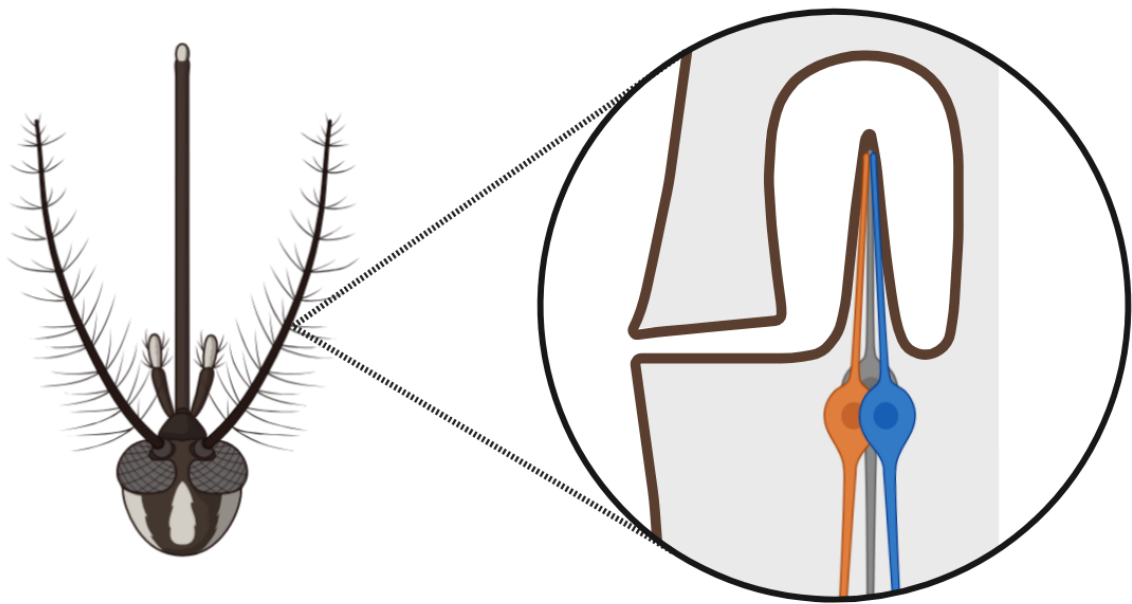


Figure 2. Mosquito antennae and hygroreceptors. *Ae. aegypti* antennae and enhanced view of hygroreceptor sensillum ampullaceum with moist (blue) and dry (orange) cells highlighted.